Housnia

Housnia – Mobile Health App Project Proposal

Menstrual health, pregnancy follow-up and postpartum care for women and girls in Chad

October 2026

Project at a glance

Item	Details
Project name	Housnia
Country	Chad (pilot), then the Sahel and Central Africa
Sector	Digital health · Sexual and reproductive health · Maternal health · Femtech
Target users	Girls aged 10–17, women aged 18–35, new mothers, community health workers
Solution	Free, offline-first mobile app with short texts in French and Arabic, and audio and video in French, Arabic and Chadian Arabic
Stage	Concept and design completed; seeking funding to build the prototype and run a pilot
Year-1 budget	114.4 million FCFA (≈ €174,400)
Founder	Housna Mahamat Ali, student midwife, University of N'Djamena
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1. Executive summary

Housnia is a free mobile application that works without an internet connection. It helps women and girls in Chad understand their periods, follow their pregnancy, and take care of themselves and their baby after childbirth. Written content is available in French and Arabic. Audio and video content is available in French, Arabic and Chadian Arabic, so that women who cannot read can still use the app.

- **Vision**: no girl in Chad should face her first period in fear, and no woman should miss a danger sign during pregnancy because she lacked information.
- **Mission**: put reliable health information, validated by Chadian midwives and doctors, in every woman's pocket, in her own language and without needing a network.
- **Name**: "Housnia" comes from the Arabic root *husn* (beauty, goodness). It is short, feminine and easy to pronounce in French, Arabic and English.

Target users

Profile	Age	Main need
Girls	10–17	Understanding puberty and first periods, menstrual hygiene
Young women	18–35	Cycle tracking, preparing for and following pregnancy
New mothers	All ages	Postpartum recovery, breastfeeding, newborn care
Community health workers	Adults	Facilitators, midwives and teachers who use the app for group awareness sessions

2. Founder

Housna Mahamat Ali is a second-year student in Obstetric Care (Midwifery) at the Faculty of Human Health Sciences, University of N'Djamena, Chad.

Her training as a future midwife puts her close to the problem Housnia addresses: she sees how many women and girls lack basic information on menstruation, pregnancy and postpartum care, and how often danger signs are recognised too late. Housnia is her answer to that gap: a tool designed in Chad, for Chadian women, by a future health professional.

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3. The problem

Chad has one of the highest maternal mortality rates in the world. Many deaths are linked to danger signs that are not recognised, or to care that is sought too late. Reproductive health information remains scarce, is often taboo, and is rarely available in Chadian Arabic, the language many women speak every day.

Indicator	Value	Source
Maternal mortality	860 deaths per 100,000 live births	WHO Chad – Health Statistics Yearbook 2022
Births attended by skilled personnel	38% (2022)	UNICEF – Chad Country Programme
Doctor density	about 0.85 per 10,000 people	WHO Chad
Literacy, age 15 and over	22.3%	CIA World Factbook
Girls married before 18	nearly 70% (estimate)	The Leprosy Mission

What this means for the design of Housnia

- **Low literacy** → all key content must exist as audio and video, with icons, not only as text.
- **Unreliable mobile network and expensive data** → the app must work fully offline.

- **Early marriage and adolescent pregnancy** → adolescent girls are a priority audience, with age-appropriate language.
- **Few health workers** → the app supports midwives and community health workers rather than replacing them, and always directs users to a health centre in case of danger.
- **Taboos** → discretion by design: PIN code, neutral app icon, no data shared without consent.

4. Objectives and indicators

The overall objective is to improve knowledge of menstrual, prenatal and postnatal health among Chadian women and girls, and to increase the use of health services. The targets below are proposals to be refined with partners.

Specific objective	Indicator	2-year target (proposed)
Reach the audience	Active installations	100,000
Inform adolescent girls	Users aged 10–17 who completed the “My Periods” module	30,000
Improve knowledge	Average quiz score before / after a module	+30 points
Encourage antenatal care	Pregnant women with antenatal visit reminders switched on	20,000
Recognise danger signs	Users able to name 3 danger signs (field survey)	70%
Build local ownership	Community health workers trained to use the app	500

Usage indicators are collected anonymously, only with the user’s consent, when the phone synchronises (see section 9).

5. Features and modules

The app has four tabs — **My Periods**, **My Pregnancy**, **After Baby**, **Learn** — plus an **Emergency** button that is always visible.

Module	Main features	For whom
My Periods	Cycle calendar, prediction of next period and fertile window, symptom diary (pain, mood, flow), discreet reminders, guides on puberty and menstrual hygiene (reusable pads, water, privacy)	Girls and women
My Pregnancy	Last-period date entry, due-date calculation, week-by-week follow-up (baby’s development, advice), reminders for the 4 to 8 antenatal visits, vaccines, nutrition, birth preparation (bag, transport, health centre)	Pregnant women
After Baby	Follow-up of the 42 days after birth, exclusive breastfeeding, cord care, newborn vaccination, family planning, mother’s mental health (fatigue, sadness)	New mothers
Learn	Library of videos and guides by	All users, community health

Module	Main features	For whom
	topic, quizzes, myths and facts, notebook of questions to ask the midwife	workers
Emergency	Illustrated danger signs (heavy bleeding, convulsions, high fever, severe headache, baby no longer moving), emergency number, advice to go to the health centre immediately	All users

Cross-cutting features

- Language choice at first launch (text language and audio/video language chosen separately), changeable at any time.
- No name or phone number required; only age and situation.
- PIN code and “neutral icon” mode for privacy.
- “Group mode” for community health workers: full-screen video playback for awareness sessions.
- Offline directory of health centres by province, updated with the content.

Version 2 (proposed): phone-to-phone content sharing without internet (Bluetooth / Wi-Fi Direct), voice-based question-and-answer assistant, and a version for fathers and husbands.

6. Content: written guides and videos

The launch includes about 60 written guides in French and Arabic, and 30 short videos (2 to 4 minutes) in French, Arabic and Chadian Arabic, all validated by a Chadian medical committee.

Guide format: a simple title, 3 to 5 short sentences, an illustration, a “Key point” box, a “Listen” button (recorded female voice, also in Chadian Arabic) and, where needed, a red box “Go to the health centre if…”.

Video format: 2D animation or live action with Chadian actresses, voice-over in French, Arabic or Chadian Arabic, subtitles in French or Arabic, 480p resolution to keep each video around 8 to 15 MB.

Topic	Examples of guides and videos
Puberty and periods	My body is changing; What are periods?; My first period; Hygiene and protection; Pain: what is normal and what is not; Myths about periods
Cycle and fertility	Understanding your cycle; Fertile window; Irregular periods
Pregnancy	Am I pregnant?; Antenatal visits; Eating well during pregnancy; Iron and folic acid; Malaria and pregnancy; Danger signs; Preparing for birth; Giving birth at the health centre
After childbirth	The 42 days after birth; Bleeding after birth; Breastfeeding; Newborn care; Baby’s vaccines; Spacing births; Sadness after birth
Rights and protection	My body belongs to me; Early marriage and health; Where to find help

Production process

1. Writing in French by a midwife and a health writer.

2. Validation by the medical committee (Ministry of Public Health, midwives, gynaecologist).
3. Translation of texts into Arabic; oral adaptation into Chadian Arabic for audio and video; testing with women in N'Djamena and one province.
4. Audio recording and video production.
5. Cultural and religious review (community leaders) for acceptability.
6. Release as a downloadable "content pack" (section 9).

7. Languages: written in 2 languages, spoken in 3

Texts are in **French** and **Arabic**. Audio and video are in **French**, **Arabic** and **Chadian Arabic**. Chadian Arabic is mainly spoken and rarely written, so it is offered only orally — which also reaches women who cannot read.

Language	Texts (guides, screens)	Audio and video	Key point
French	Yes	Yes	Simple vocabulary, no unexplained medical terms
Arabic	Yes	Yes	Fully mirrored, right-to-left interface
Chadian Arabic	No	Yes	Native female voices, mandatory field testing

- At first launch, the user chooses the text language (French or Arabic) and the audio/video language (French, Arabic or Chadian Arabic), using large buttons with each language name spoken aloud.
- No content is published until it exists in all required languages; no half-translated screens.
- Sensitive terms (periods, childbirth, body parts) are chosen with Chadian women so they are understood without causing offence.
- More languages can be added later (Sara, Kanembu…) using the same structure.

8. Visual identity

Pink is the main colour, paired with a deep plum for readability, a warm sand tone that recalls the Sahel, and a calming teal for health. Each main module has its own colour, which helps users with low literacy find their way.



Housnia colour palette



Housnia logo

Logo: a stylised female figure made of a crescent and a drop, in pink on a sand background, with the name “Housnia” alone in Latin letters, like major international apps. The plum crescent evokes the veil and protection, the pink drop the female body and the cycle, and the small light drop at the centre the life that grows.

9. Offline operation and updates

Once installed, the whole app works without a network: texts, audio, calendars, reminders and personal data are stored on the phone. The network is only used to receive new content.

Target size

- Core app (interface in 2 languages, all guides, compressed audio in 3 languages): about 40 MB.
- Videos: downloaded by topic pack (about 50 to 100 MB each), as the user chooses, or copied from an SD card or a community health worker’s phone.

Progressive updates

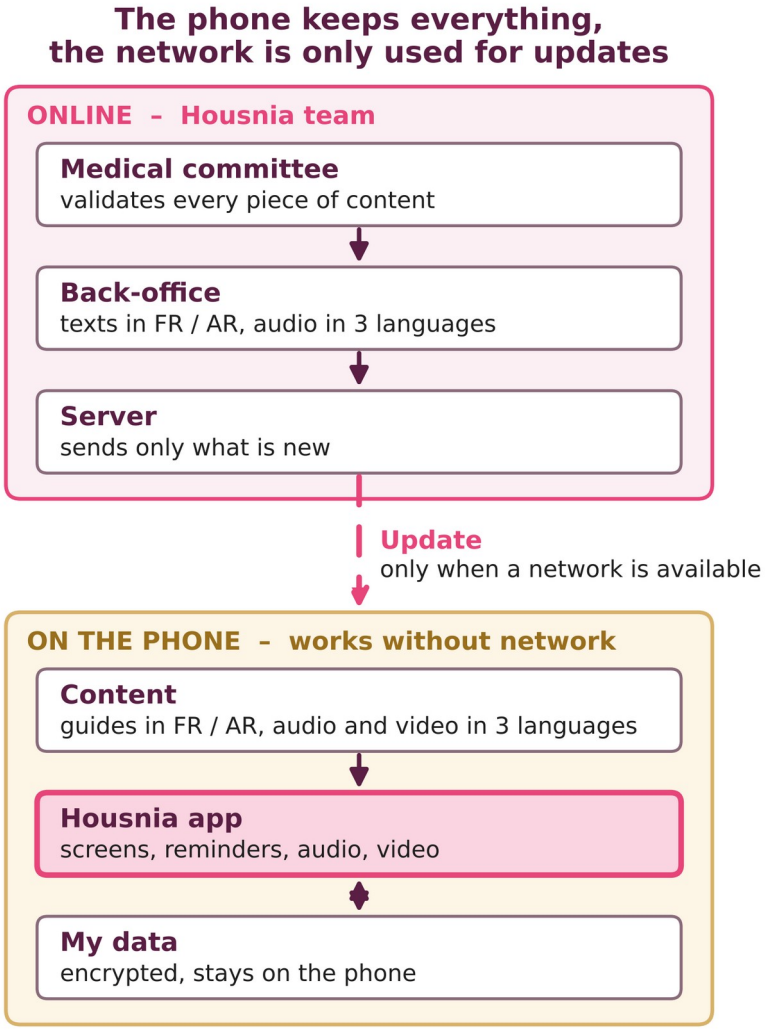
1. Content is split into versioned “packs” (by topic and language).
2. When the phone has internet access (preferably Wi-Fi), the app quietly checks for new packs.
3. Only new or changed items are downloaded, in the background, resuming if the connection drops.
4. New content is activated only once complete and verified; the previous version stays usable until then.
5. A “New content available” message appears in the Learn tab.

Other offline distribution channels

- Installation file (APK) shared by Bluetooth, ShareIt or SD card.
- “Housnia Points” in health centres, schools and associations: a computer or local Wi-Fi hotspot to install and update the app.
- Community health workers receive updates first and pass them on.

10. Technical architecture and privacy

The app is built with Flutter, first for Android (the large majority of phones in Chad), with all content and data stored on the phone.



Housnia architecture: the phone keeps everything, the network is only used for updates

Component	Proposed choice
App	Flutter (Android 6+, phones with 1–2 GB RAM), iOS later
Local storage	Encrypted SQLite (SQLCipher) for personal data; audio/video files in app storage
Reminders	Local notifications, no server needed
Translations	Separate language files, full right-to-left support
Back-office	Open-source headless CMS (Strapi or Directus)
Content packs	Versioned, signed archives with resumable download
Analytics	Anonymous, queued locally, sent only with consent
Licence	Open-source code (e.g. MIT or GPL), reusable by other countries and partners

Privacy

- No account, no name, no phone number required.

- Cycle, pregnancy and health data never leave the phone.
- PIN code, neutral icon and an “erase all my data” button.
- Compliance with Chad’s personal data protection law, to be confirmed with a lawyer.

11. Implementation plan

The first public version can be released about 9 months after the start, then enriched with new content packs every 2 to 3 months.

Phase	Timing	Deliverables
1. Scoping and field research	Months 1–2	Interviews with women, girls and midwives; topic validation; partnership with the Ministry of Health
2. Design	Months 2–3	Brand guidelines, mock-ups, testing of icons and user flow
3. Content	Months 2–7	60 guides in 2 languages, 30 videos and audio in 3 languages, medical validation
4. Development	Months 3–7	Offline Android app, content server
5. Pilot	Month 8	Test with 300 users in N’Djamena and 1–2 provinces
6. Launch	Month 9	Release on Google Play and as an APK, training of community health workers
7. Updates	Ongoing	New content packs, iOS version if needed, new languages

Team to be built

- Founder and project lead: Housna Mahamat Ali
- 2 mobile developers (Flutter) and 1 back-end developer
- 1 UI/UX designer and 1 illustrator
- Medical committee: midwives, gynaecologist, paediatrician
- Translators (French, Arabic) and voice artists (French, Arabic, Chadian Arabic)
- Video team (animation or filming)
- Field coordinator and community health workers

12. Budget and funding request

Year-1 budget. Figures are estimates to be refined with local quotes. Euro amounts use the fixed rate 1 € = 655.957 FCFA.

Item	FCFA	EUR (≈)
Field research and user testing	8,000,000	12,200
Design and illustrations	10,000,000	15,200
App and server development	35,000,000	53,400
Writing, medical validation, translations	12,000,000	18,300
Audio and video production	25,000,000	38,100
Hosting, maintenance, app store	4,000,000	6,100

Item	FCFA	EUR (≈)
publication		
Community health worker training and launch	10,000,000	15,200
Contingency (10%)	10,400,000	15,900
Total	**114,400,000**	**≈ 174,400**

How funding would be used in phases

- **Phase A – Prototype (≈ US$100,000)**: field research, design, first 20 guides and 10 videos, working offline Android prototype, pilot with 300 users. This phase matches the size of seed grants such as UNICEF Femtech Ventures.
- **Phase B – National launch (remaining budget)**: full content, training of 500 community health workers, national release.

13. Scaling beyond Chad

Housnia is designed to scale. Content is organised in language and country packs, so adding a country means adding packs, not rebuilding the app.

Stage	Countries	Languages to add (audio and video)
1. Chad (years 1–2)	Chad	Chadian Arabic, then Sara and Kanembu
2. Sahel and Central Africa (years 2–4)	Niger, Mali, Burkina Faso, Cameroon, Central African Republic, Sudan, Mauritania	Hausa, Fulfulde, Bambara, Sudanese Arabic…
3. Africa and beyond (year 5+)	East Africa, South Asia, other low-resource settings	English, Swahili and others as needed

For each new country: local medical validation, national health-centre directory and emergency numbers, recordings by local women, and compliance with local data protection law.

What makes Housnia different: most period and pregnancy apps need internet access and assume users can read. Housnia works offline, speaks local languages, and serves women who cannot read — a gap that remains largely unfilled worldwide.

14. Partners and risks

Partners to approach: Ministry of Public Health and Prevention, UNFPA, UNICEF, WHO Chad, maternal health NGOs (e.g. Ensemble Sawa, Women Forward Chad, Women's Hope International), women's and youth associations, schools, and mobile operators (Airtel, Moov Africa) for zero-rated downloads.

Risk	Mitigation
Cultural or religious rejection of the topic	Review by community and religious leaders, respectful tone, involvement of families
Limited smartphone access	Group mode for community health workers, offline sharing, shared phone with PIN
Medical error in content	Mandatory medical committee validation before each pack
Women who read neither French nor Arabic	All key content available as audio and video, including in Chadian Arabic
Privacy on shared phones	PIN, neutral icon, data stored only on the phone

Risk	Mitigation
Funding after year 1	Institutional partnerships, mobile operator sponsorship, grant calls

Contact

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